



Deploy an App with Docker



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```
jerry@DESKTOP-1ESWKL:~$ docker
Usage: docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:
run          Create and run a new container from an image
exec        Execute a command in a running container
ps          List containers
build       Build an image from a Dockerfile
bake        Build from a file
pull        Download an image from a registry
push        Upload an image to a registry
images      List images
login        Authenticate to a registry
logout      Log out from a registry
search      Search Docker Hub for images
version     Show the Docker version information
info        Display system-wide information

Management Commands:
builder      Manage builds
buildx*      Docker Buildx
compose*     Docker Compose
container    Manage containers
context      Manage contexts
image        Manage images
manifest     Manage Docker image manifests and manifest lists
network      Manage networks
plugin       Manage plugins
system       Manage Docker
volume       Manage volumes

Swarm Commands:
swarm        Manage Swarm

Commands:
attach       Attach local standard input, output, and error streams to a running container
commit       Create a new image from a container's changes
cp           Copy files/folders between a container and the local filesystem
create       Create a new container
diff         Inspect changes to files or directories on a container's filesystem
events       Get real time events from the server
export       Export a container's filesystem as a tar archive
history      Show the history of an image
import       Import the contents from a tarball to create a filesystem image
inspect      Return low-level information on Docker objects
kill         Kill one or more running containers
load         Load an image from a tar archive or STDIN
logs         Fetch the logs of a container
pause        Pause all processes within one or more containers
port         List port mappings or a specific mapping for the container
rename       Rename a container
restart      Restart one or more containers
rm           Remove one or more containers
rmi          Remove one or more images
save         Save one or more images to a tar archive (streamed to STDOUT by default)
start        Start one or more stopped containers
stats        Display a live stream of container(s) resource usage statistics
stop         Stop one or more running containers
tag          Create a tag TARGET_IMAGE that refers to SOURCE_IMAGE
top          Display the running processes of a container
unpause      Unpause all processes within one or more containers
```



Introducing Today's Project!

What is Docker?

Docker is an open-source platform designed to simplify the development, deployment, and management of applications using containerization. Containers are lightweight, portable, and isolated environments that package an application along with its dependencies, ensuring consistent behavior across different systems. I used Docker in this project to demonstrate how to create and deploy an app using Docker

One thing I didn't expect...

One thing I didn't expect in this project was not really hard to understand.

This project took me...

This project took me approximately an hour. The most challenging part was deploying in AWS Elastic Beanstalk. It was most rewarding to know how to create a containerized app using docker and to know how to deploy it to cloud using AWS

A circular profile picture of a young man with light blue hair and a red bow tie.

Understanding Containers and Docker

Containers

Containers are small VMs where the image runs. They are useful because they make it easy to deploy to other devices

A container image is a piece of instruction of what to build inside the container

Docker

Docker is an open-source platform designed to simplify the development, deployment, and management of applications using containerization. Docker Desktop is a UI for Docker to manage images and containers easily

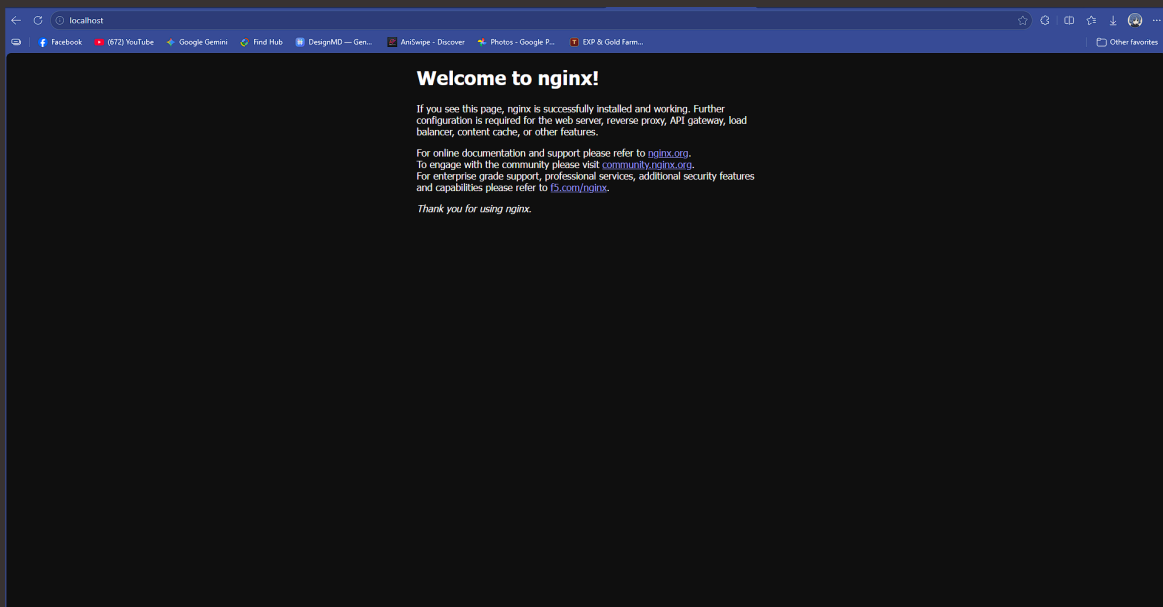
The Docker daemon is a service that runs on your host operating system and manages Docker containers.



Running an Nginx Image

Nginx is a web server, which is a program you use to run websites and web apps.

The command I ran to start a new container was `docker run -d -p 80:80 nginx -d` for detach -p for publish ports - 80(internal port):80(external port) - nginx is the image to run in a container

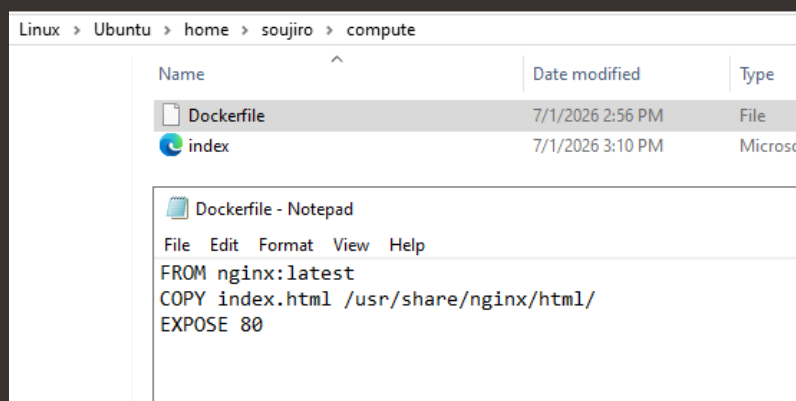


Creating a Custom Image

A Dockerfile is a text file that contains a set of instructions used to build a Docker image.

The first line, FROM nginx:latest, means our image starts as a copy of the latest Nginx image COPY index.html /usr/share/nginx/html/ replaces the default HTML file provided by Nginx with my own custom index.html file EXPOSE 80 means I want the container to receive web traffic through port 80

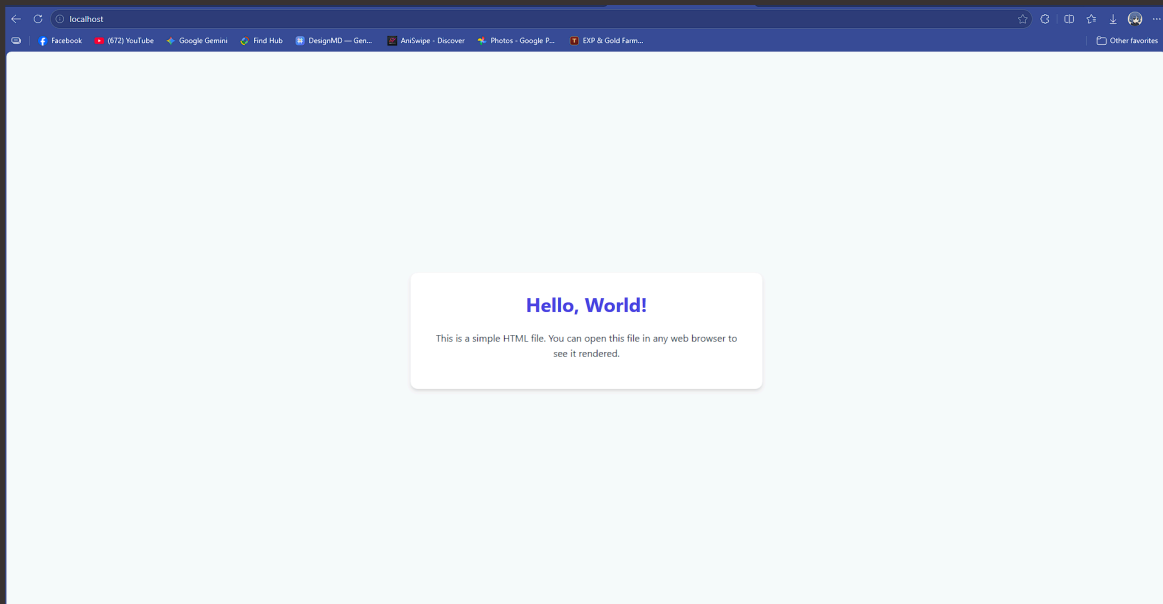
The command I used to build a custom image with my Dockerfile was docker run -t my-web-app . The '.' at the end of the command means combine all instruction applied to image together



Running My Custom Image

There was an error when I ran my custom image because the port is use by other container I resolved this by stopping the container that using the port 80

In this example, the container image is the blueprint that tells Docker the application code, dependencies, libraries, etc that should go into a container. The container is the actual software that's created from this image and running the web server displaying your index.html

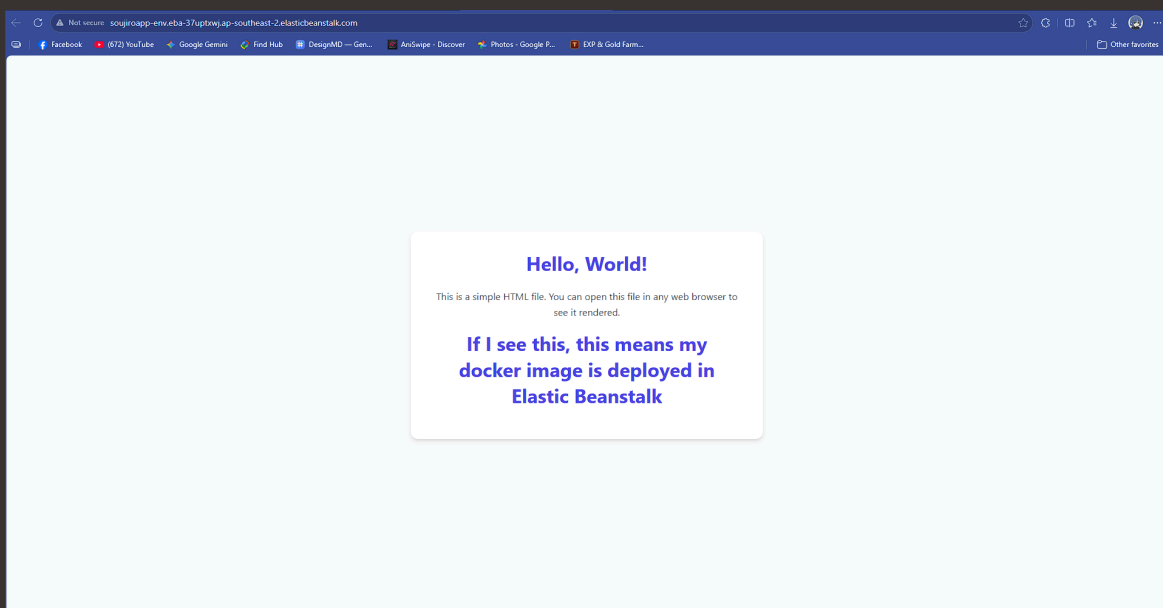




Elastic Beanstalk

Elastic Beanstalk is AWS's way of running your Docker containers without the setup headaches. Build and test your app locally, wrap it into a Docker image, then hand that image to Elastic Beanstalk. It spins up the servers, launches your container.

Deploying my custom image with Elastic Beanstalk took me less than 5 minutes

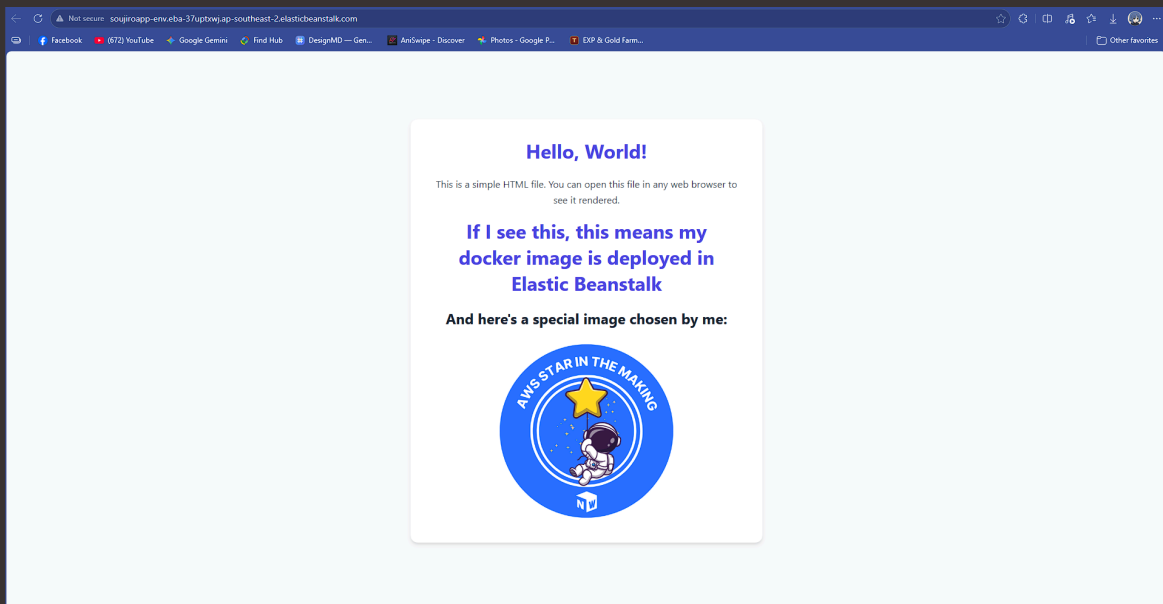




Deploying App Updates

To learn how to deploy app updates with Elastic Beanstalk, I updated my app by adding new heading and image, I verified those changes by opening the HTML file to the browser by right clicking the file

My app updates didn't show up in my live environment immediately because it was not updated yet to the AWS. To deploy my changes, I only had to upload my new files and deploy it.





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